ATKINS A member of the WS Atkins Group **PO Box 5620** **Dubai, United Arab Emirates**	Project GL AL QUSAIS DEPOT (Bldg 9)			Job ref.	
	Part of Structure RESERVOIR'S STAIR			Calc. sheet no. /	rev. /
	Drawing ref.	Calc. by COC	Date 11 MAY '09	Check by	Date

Ref.	Calculations	Output

DESIGN OF STAIR

Live load = 5 kPa

Dead load = 0.5 kPa

$W_{LL} = 5(1.1/2) = 2.8$ kN/m

$W_{DL} = 0.5(1.1/2) = 0.3$ kN/m

- for member design checking refer to p. 28

- solving for base plate design at mid-landing;

$V = 0.1W = 0.1(25) = 2.5$ kN (conservative)

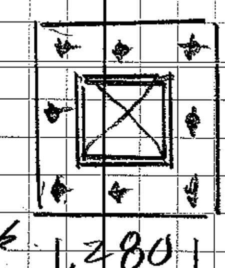
$M = 2.5(h) = 2.5(2.2) = 6.3$ kN-m

- assuming 50% of bolt will carry moment only;

$M = Te \rightarrow 6.3 = T(0.28)$

$T = 23$ kN / 3 bolts

$= 8$ kN



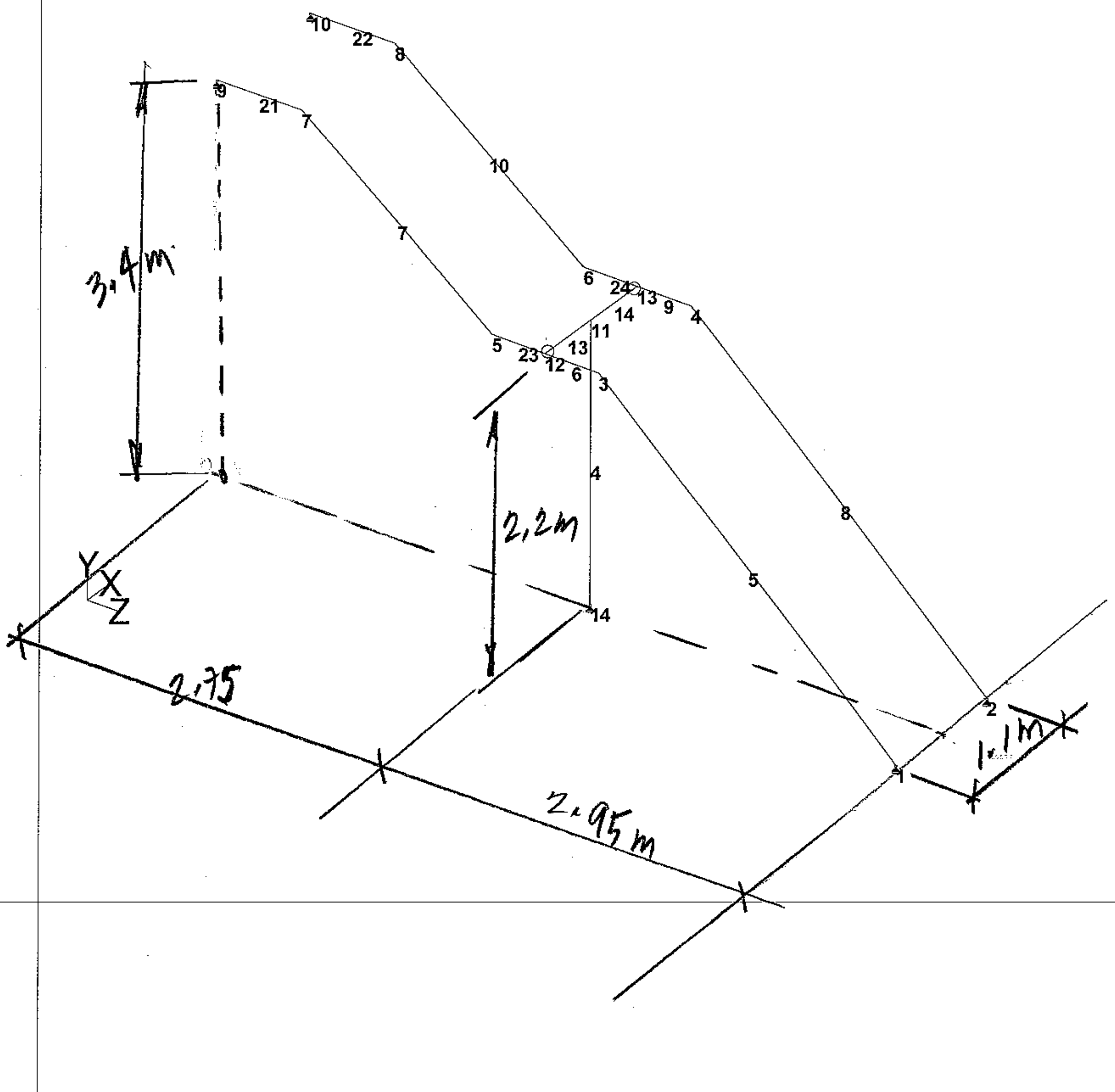
(from p. 29) $T_{cap} = 0.8(12.1)$

$= 9.7$ kN > 8 kN ok!

∴ Use 8 M16 (G5.8) ANCHOR BOLTS WITH (24) 20 mm THK BASE PLATE.

GL AL QUSAIS DEPOT (Bldg 9)	Job No	Sheet No 1	Rev
	Part		
Job Title RESERVOIR'S STAIR	Ref		
	By CBC	Date	Chd
Client	File reservoir STAIR_1.std	Date/Time 11-May-2009 06:43	

STRUCTURAL MODEL (STAIR)
(JOINT AND MEMBER NUMBERING)



```
**************************************************
*                                                *
*           STAAD.Pro                            *
*           Version  2007    Build 02            *
*           Proprietary Program of               *
*           Research Engineers,  Intl.           *
*           Date=     MAY 11, 2009               *
*           Time=      6:44:18                   *
*                                                *
*      USER ID: WS Atkins and Partners Overseas  *
**************************************************
```

```
INPUT FILE: reservoir STAIR_1.STD
     1. STAAD SPACE STAIR CASE
     2. START JOB INFORMATION
     3. ENGINEER DATE 11-OCT-07
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 0.15 0 0; 2 1.25 0 0; 3 0.15 2.2 -2.5; 4 1.25 2.2 -2.5; 5 0.15 2.2 -3.4
     9. 6 1.25 2.2 -3.4; 7 0.15 3.4 -5; 8 1.25 3.4 -5; 9 0.15 3.4 -5.7
    10. 10 1.25 3.4 -5.7; 11 0.7 2.2 -2.95; 12 0.15 2.2 -2.95; 13 1.25 2.2 -2.95
    11. 14 0.7 0 -2.95
    12. MEMBER INCIDENCES
    13. 4 14 11; 5 1 3; 6 3 12; 7 5 7; 8 2 4; 9 4 13; 10 6 8; 13 12 11; 14 11 13
    14. 21 7 9; 22 8 10; 23 5 12; 24 6 13
    15. DEFINE MATERIAL START
    16. ISOTROPIC STEEL
    17. E 2.05E+008
    18. POISSON 0.3
    19. DENSITY 76.8195
    20. ALPHA 1.2E-005
    21. DAMP 0.03
    22. END DEFINE MATERIAL
    23. CONSTANTS
    24. BETA 180 MEMB 5 TO 7 13 21 24
    25. MATERIAL STEEL ALL
    26. MEMBER PROPERTY BRITISH
    27. 4 13 14 TABLE ST TUBE TH 0.006 WT 0.2 DT 0.2
    28. 5 TO 10 21 TO 24 TABLE ST CH230X75X26
    29. SUPPORTS
    30. 1 2 9 10 14 PINNED
    31. MEMBER RELEASE
    32. 13 START MY MZ
    33. 14 END MY MZ
    34. LOAD 1 DL
    35. SELFWEIGHT Y -1
    36. LOAD 2 SDL
    37. MEMBER LOAD
    38. 5 TO 10 21 TO 24 UNI GY -0.3
    39. LOAD 3 LL
    40. MEMBER LOAD
```



```
    41. 5 TO 10 21 TO 24 UNI GY -2.8
    42. LOAD COMB 4 SLS
    43. 1 1.0 2 1.0 3 1.0
    44. LOAD COMB 5 ULS
    45. 1 1.4 2 1.4 3 1.6
    46. PERFORM ANALYSIS
```

```
         P R O B L E M   S T A T I S T I C S
         -----------------------------------

   NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS =     14/     13/      5
   ORIGINAL/FINAL BAND-WIDTH=      9/      3/     24 DOF
```

```
    TOTAL PRIMARY LOAD CASES =     3, TOTAL DEGREES OF FREEDOM =      69
    SIZE OF STIFFNESS MATRIX =        2 DOUBLE  KILO-WORDS
    REQRD/AVAIL. DISK SPACE  =     12.0/  6725.9 MB,  EXMEM = 1214.5 MB



   47. LOAD LIST 4
   48. PRINT SECTION DISPL LIST 5 7
 SECTION  DISPL    LIST     5
     STAIR CASE                                                -- PAGE NO.    3



    MEMBER SECTION DISPLACEMENTS
    ----------------------------
    UNIT =INCHES FOR FPS AND CM FOR METRICS/SI SYSTEM

    MEMB   LOAD     GLOBAL X,Y,Z DISPL FROM START TO END JOINTS AT 1/12TH PTS

      5      4        0.0000    0.0000    0.0000    0.0000   -0.0108   -0.0090
                      0.0000   -0.0207   -0.0173    0.0000   -0.0291   -0.0242
                      0.0000   -0.0355   -0.0293    0.0000   -0.0395   -0.0324
                      0.0000   -0.0412   -0.0334    0.0000   -0.0406   -0.0324
                      0.0000   -0.0380   -0.0297    0.0000   -0.0339   -0.0256
                      0.0000   -0.0291   -0.0209    0.0000   -0.0244   -0.0163
                      0.0000   -0.0210   -0.0128

    MAX LOCAL  DISP =     0.04100    AT      138.76  LOAD    4   L/DISP=    8123

      7      4        0.0000   -0.0301   -0.0119    0.0000   -0.0324   -0.0134
                      0.0000   -0.0342   -0.0145    0.0000   -0.0354   -0.0151
                      0.0000   -0.0361   -0.0153    0.0000   -0.0360   -0.0150
                      0.0000   -0.0353   -0.0142    0.0000   -0.0339   -0.0129
                      0.0000   -0.0318   -0.0111    0.0000   -0.0292   -0.0089
                      0.0000   -0.0262   -0.0064    0.0000   -0.0229   -0.0036
                      0.0000   -0.0193   -0.0007

    MAX LOCAL  DISP =     0.01316    AT      100.00  LOAD    4   L/DISP=  15193

        ************ END OF SECT DISPL RESULTS ***********

   49. PRINT JOINT DISPLACEMENT LIST 12
 JOINT     DISPLACE LIST      12
     STAIR CASE                                                -- PAGE NO.    4



  JOINT DISPLACEMENT (CM   RADIANS)     STRUCTURE TYPE = SPACE
  ------------------

 JOINT  LOAD   X-TRANS   Y-TRANS   Z-TRANS   X-ROTAN   Y-ROTAN   Z-ROTAN


    12    4     0.0000   -0.0196   -0.0124   -0.0001    0.0000    0.0000


  ************** END OF LATEST ANALYSIS RESULT **************


   50. PRINT SUPPORT REACTION
 SUPPORT  REACTION
     STAIR CASE                                                -- PAGE NO.    5



  SUPPORT REACTIONS -UNIT KN    METE     STRUCTURE TYPE = SPACE
  -----------------

 JOINT  LOAD   FORCE-X   FORCE-Y   FORCE-Z     MOM-X     MOM-Y     MOM Z
```

1	4	0.00	10.64	-6.87	0.00	0.00	0.00
2	4	0.00	10.64	-6.87	0.00	0.00	0.00
9	4	0.00	1.07	6.91	0.00	0.00	0.00
10	4	0.00	1.07	6.91	0.00	0.00	0.00
14	4	0.00	24.20	-0.08	0.00	0.00	0.00

************** END OF LATEST ANALYSIS RESULT **************

```
   51. LOAD LIST 5
   52. PARAMETER
   53. CODE BS5950
   54. CHECK CODE ALL
STEEL DESIGN
```

STAAD.Pro CODE CHECKING - (BSI)
****

PROGRAM CODE REVISION V2.9_5950-1_2000
STAIR CASE -- PAGE NO. 6

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
4	ST TUB E	PASS	BS-4.7 (C)	0.030	5
		37.74 C	0.00	0.00	0.00
5	ST CH2 30X75X26	PASS	BS-4.8.3.3.1	0.180	5
		18.13 C	0.00	3.85	-
6	ST CH2 30X75X26	PASS	BS-4.8.3.3.1	0.086	5
		10.76 C	0.00	4.75	-
7	ST CH2 30X75X26	PASS	BS-4.8.3.3.1	0.086	5
		16.16 C	0.00	2.74	-
8	ST CH2 30X75X26	PASS	BS-4.8.3.3.1	0.180	5
		18.13 C	0.00	3.85	-
9	ST CH2 30X75X26	PASS	BS-4.8.3.3.1	0.086	5
		10.76 C	0.00	4.75	-
10	ST CH2 30X75X26	PASS	BS-4.8.3.3.1	0.086	5
		16.16 C	0.00	2.74	-
13	ST TUB E	PASS	BS-4.8.3.3.3	0.108	5
		0.00	0.03	10.00	-
14	ST TUB E	PASS	BS-4.8.3.3.3	0.108	5
		0.00	0.03	10.00	-
21	ST CH2 30X75X26	PASS	BS-4.8.3.3.1	0.019	5
		10.82 C	0.00	0.27	-
22	ST CH2 30X75X26	PASS	BS-4.8.3.3.1	0.019	5
		10.82 C	0.00	0.27	-
23	ST CH2 30X75X26	PASS	BS-4.8.3.3.1	0.089	5
		10.82 C	0.00	4.88	-
24	ST CH2 30X75X26	PASS	BS-4.8.3.3.1	0.089	5
		10.82 C	0.00	4.88	-

************** END OF TABULATED RESULT OF DESIGN **************

55. FINISH

*********** END OF THE STAAD.Pro RUN ***********

**** DATE= MAY 11,2009 TIME= 6:44:19 ****

Product Information

Hilti HIT HY 150 injection technique
HAS, HAS-R, HAS-HCR

Recommended loads

The recommended loads are only valid for TE drilling machine drilled holes.

Anchor rod HAS, HAS R, HAS HCR
Recommended loads F_{30} in kN, concrete f_{cc} = 30 N/mm²

Type of loading		M 8	M 10	M 12	M 16	M 20	M 24[2]
Tensile load	0°	4.8	6.3	9.5	12.1	20.0	25.0
Combined load	30°	4.5	6.2	9.5	12.5	20.6	27.3
	45°	4.4	6.2	9.5	12.7	20.9	28.5
	60°	4.3	6.1	9.4	12.9	21.1	29.7
Shear load[1]	90°	4.0	6.0	9.4	13.3	21.7	32.0

[1] **For group fastenings the shear load values have to be multiplied with the factor 0.8.**
[2] **For the anchor rod HAS HCR M24 the recommended shear load is 26.8 kN.**

Note: The indicated, recommended load values require a careful cleaning of the hole with brushes and blow out pump. This holds true for all base materials and for anchor rods HAS as well as for internal thread sleeves HIS.

Recommended load for specific application: HAS, HAS R, HAS HCR

$$F_{rec} = F_{30} \times f_B \times f_T \times f_A \times f_R$$

For shear load $f_T = 1$

Influence of concrete strenght HAS, HAS R, HAS HCR

$$f_B = 1 + 0.01 \times (1 - \frac{\alpha}{90})(f_{cc,act} - 30)\ (20 \le f_{cc,act} \le 55)$$

Influence of depth of embedment HAS, HAS R, HAS HCR

$$f_T = \frac{h_{act}}{h_{nom}} \quad \text{Limiting depth of embedment } h_{lim} = 2.0\ h_{nom}$$

h_{act} = actual embedment depth

Embedment depth

HAS	M 8	M 10	M 12	M 16	M 20	M 24
h_{nom} (mm)	80	90	110	125	170	210